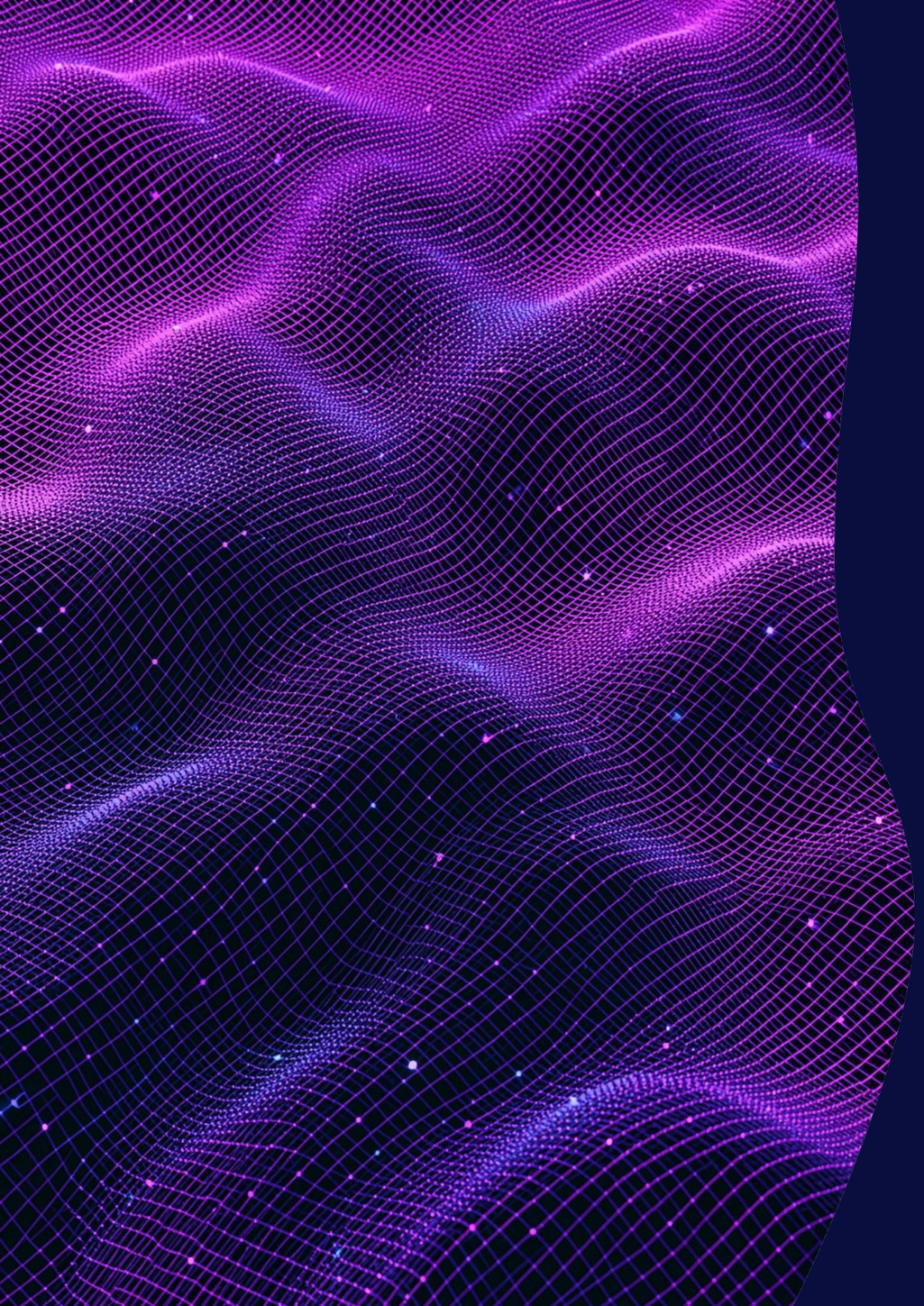

A Multi-Phase Deep Learning Framework for Botnet Detection in IoT

Novel Feature Extraction and Adaptive Classification Algorithms

PATENT DISCLOSURE · SR UNIVERSITY, INDIA

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The Problem: IoT Under Attack

Resource Heterogeneity

Varied hardware makes uniform detection policies ineffective.

Concept Drift

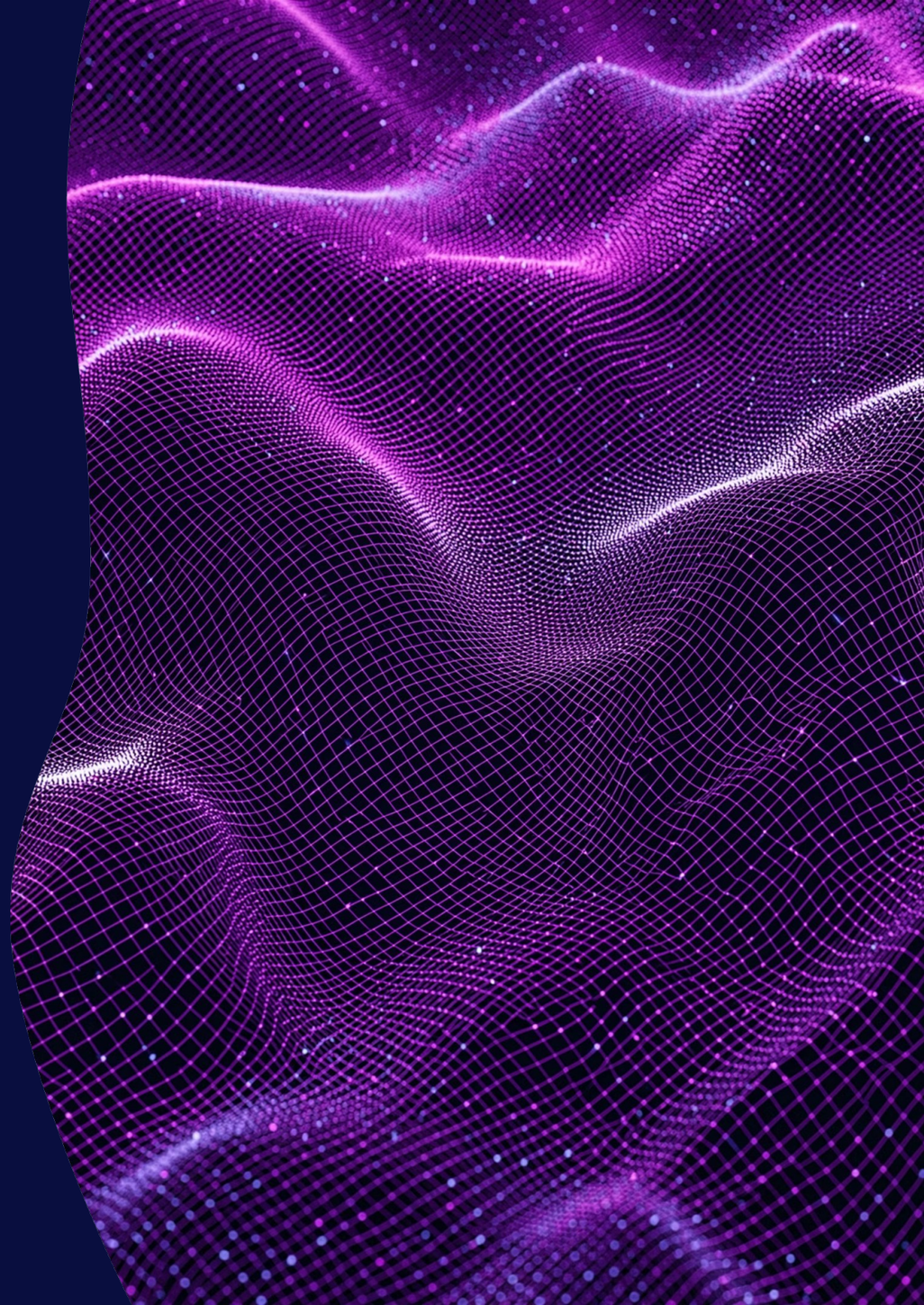
Botnets evolve C&C patterns, rendering static models obsolete.

Zero-Day Threats

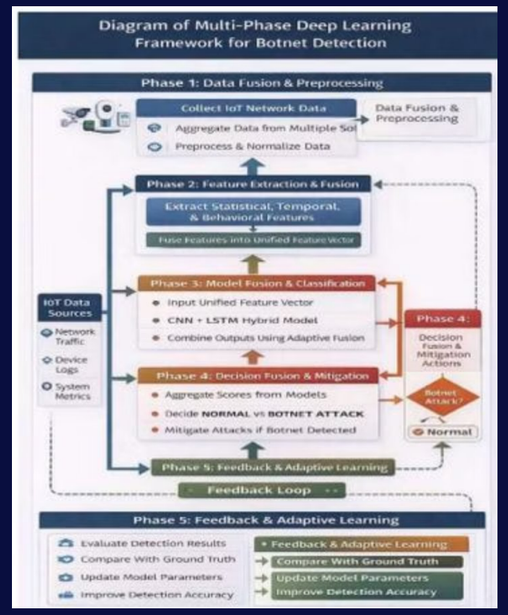
Polymorphic botnets evade signature-based and shallow detectors.

High False Positives

Existing IDS flags legitimate IoT traffic at unacceptable rates.



The Framework: Three-Phase Architecture



Phase 1 – Feature Extraction

Statistical, entropy-based, and graph-theoretic descriptors fused into a 128-D vector.

Phase 2 – Ensemble Classification

CNN, LSTM, and BiGRU branches combined via a gradient-boosted meta-learner.

Phase 3 – Adaptive Threshold

Bayesian sliding-window optimizer + EWC incremental updater for real-time adaptation.

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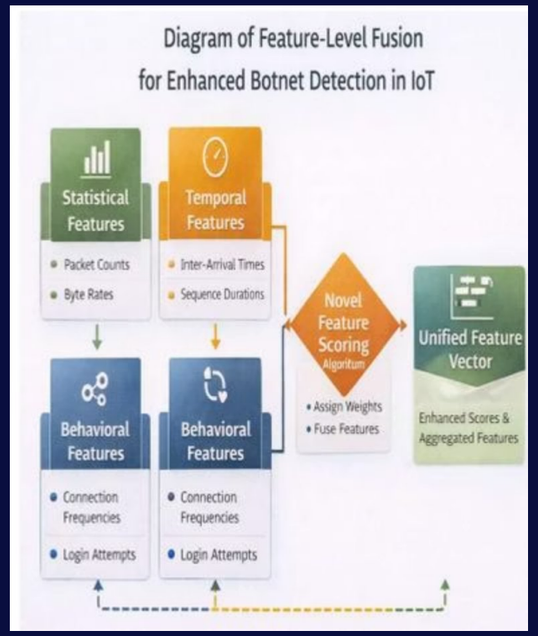
Feature Extraction & Fusion Pipeline



Three-Module System → 128-D Vector

Statistical Module

32 flow-level features: inter-arrival time, packet length, byte rate.

Entropy Module

Shannon & Rényi entropy detect encrypted C&C payload randomness.

Graph-Theoretic Module

Directed communication graph G=(V,E); derives centrality, PageRank, clustering coefficients.

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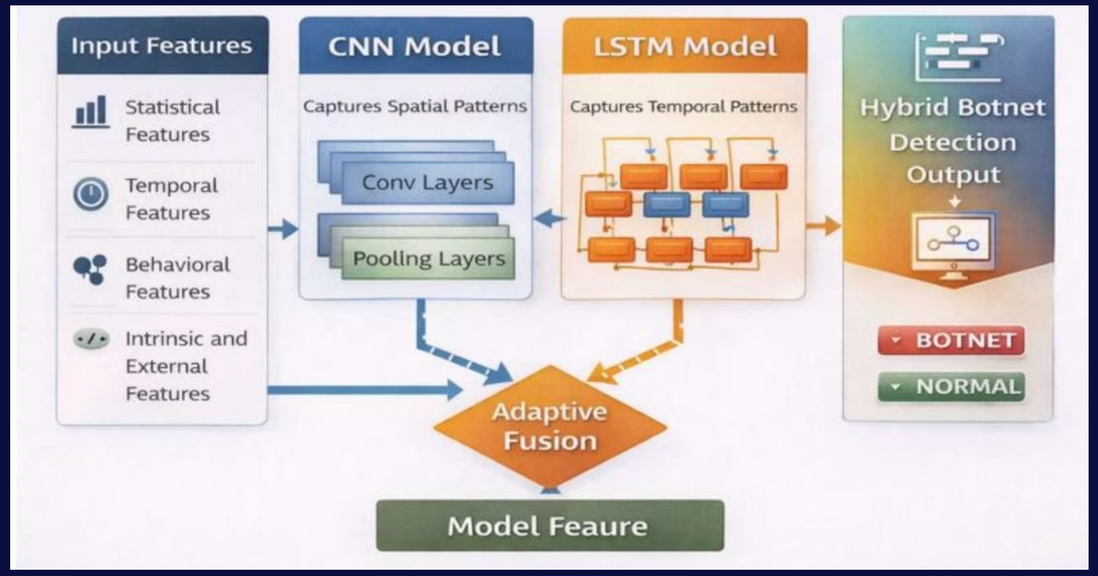
Adaptive Ensemble Classification



CNN Branch

1-D Conv with kernel sizes 3, 5, 7 — captures local spatial patterns.

LSTM Branch

2-layer stacked LSTM (256 units) models temporal flow dependencies.

BiGRU Branch

Bidirectional GRU captures asymmetric forward/backward traffic patterns.

Meta-Learner

Gradient-boosted decision tree fuses all three branch predictions.

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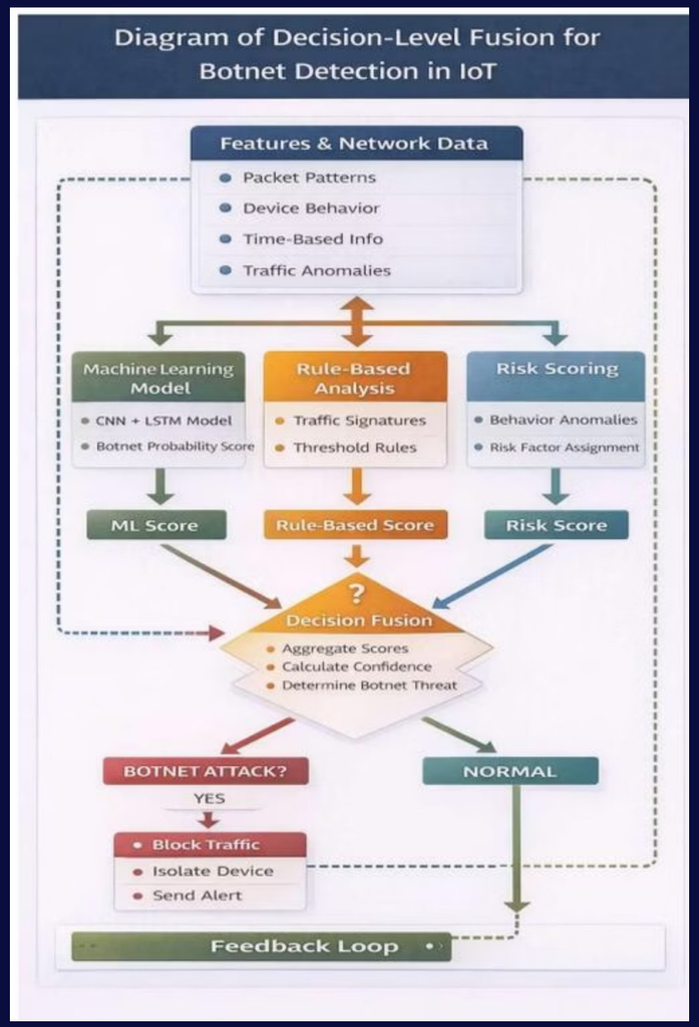
Decision Fusion & Mitigation



Three parallel scoring subsystems converge at a **Decision Fusion node**:

ML Score

CNN + LSTM botnet probability output.

Rule-Based Score

Traffic signatures and threshold rules.

Risk Score

Behavior anomalies and risk factor assignment.

⚠ On detection: Block Traffic → Isolate Device → Send SIEM Alert

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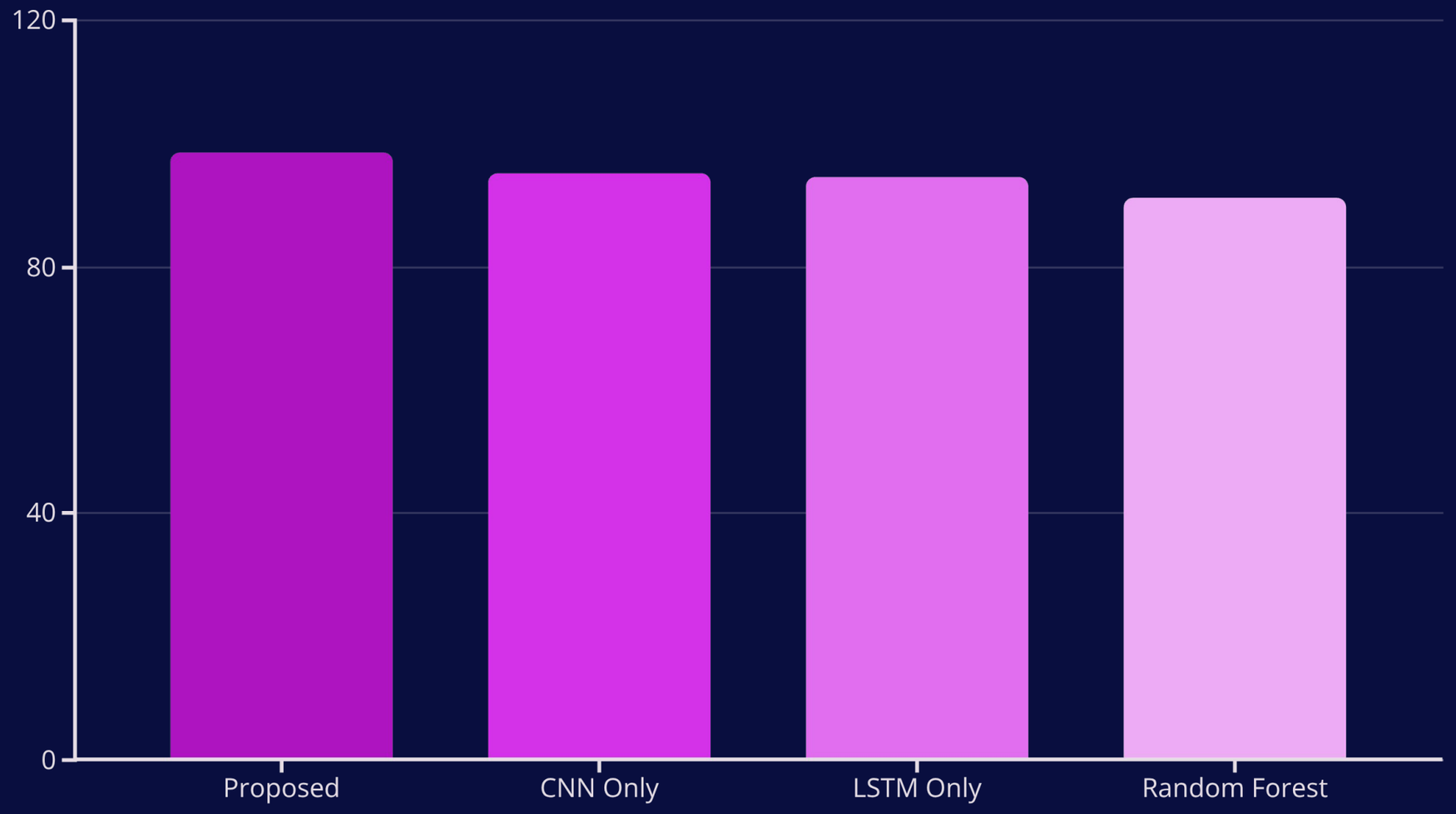
Performance Results



98.7%

Detection Accuracy

On N-BaloT, MQTTset & Bot-IoT benchmarks.

42%

Fewer False Positives

vs. state-of-the-art CNN-only classifiers.

91%

Zero-Day Detection

Of novel botnet families in blind evaluation.

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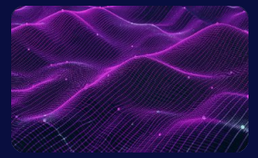
Key Advantages & Deployment

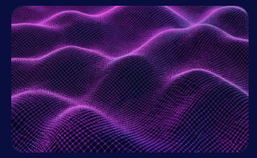


Lightweight Deployment

Quantized model runs within **256 MB RAM** — fits Raspberry Pi 4 and IoT gateways.

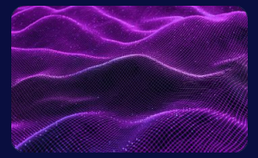
Real-Time Processing

Processes **10,000 flow records/sec** on standard edge hardware.



Adaptive Learning

EWC incremental updater reduces model update latency by **67%** without full retraining.



Explainability

Integrated SHAP layer provides human-interpretable feature attribution per detection decision.

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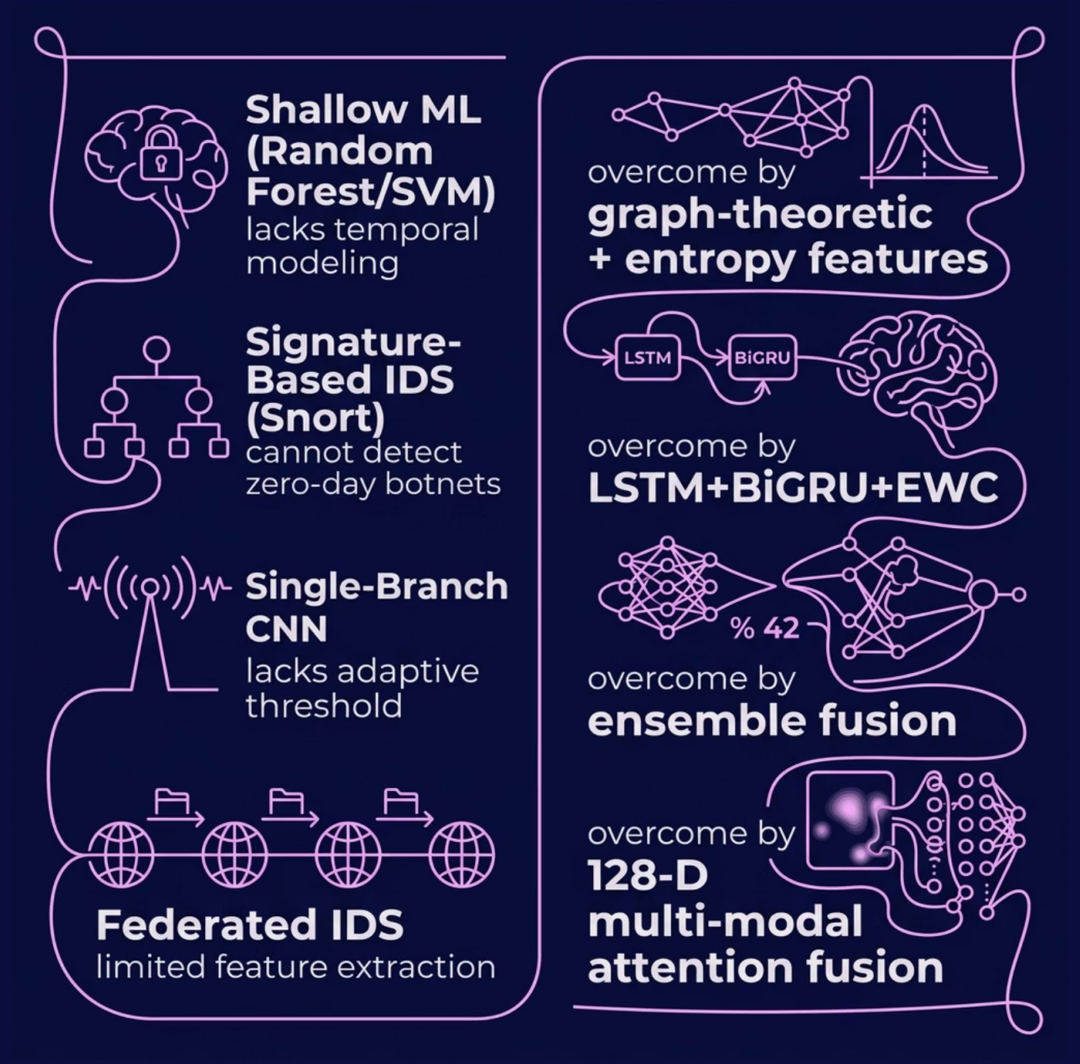
Novel Contributions vs. Prior Art



Six Novel Firsts

01

Graph-theoretic IoT C&C topology fingerprinting

02

CNN + LSTM + BiGRU with gradient-boosted meta-learner

03

Rényi entropy for encrypted C&C payload detection

04

EWC-based incremental learning in IoT security

05

Bayesian sliding-window threshold calibration

06

Unified 128-D attention-weighted multi-modal fusion

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Patent Status & Next Steps

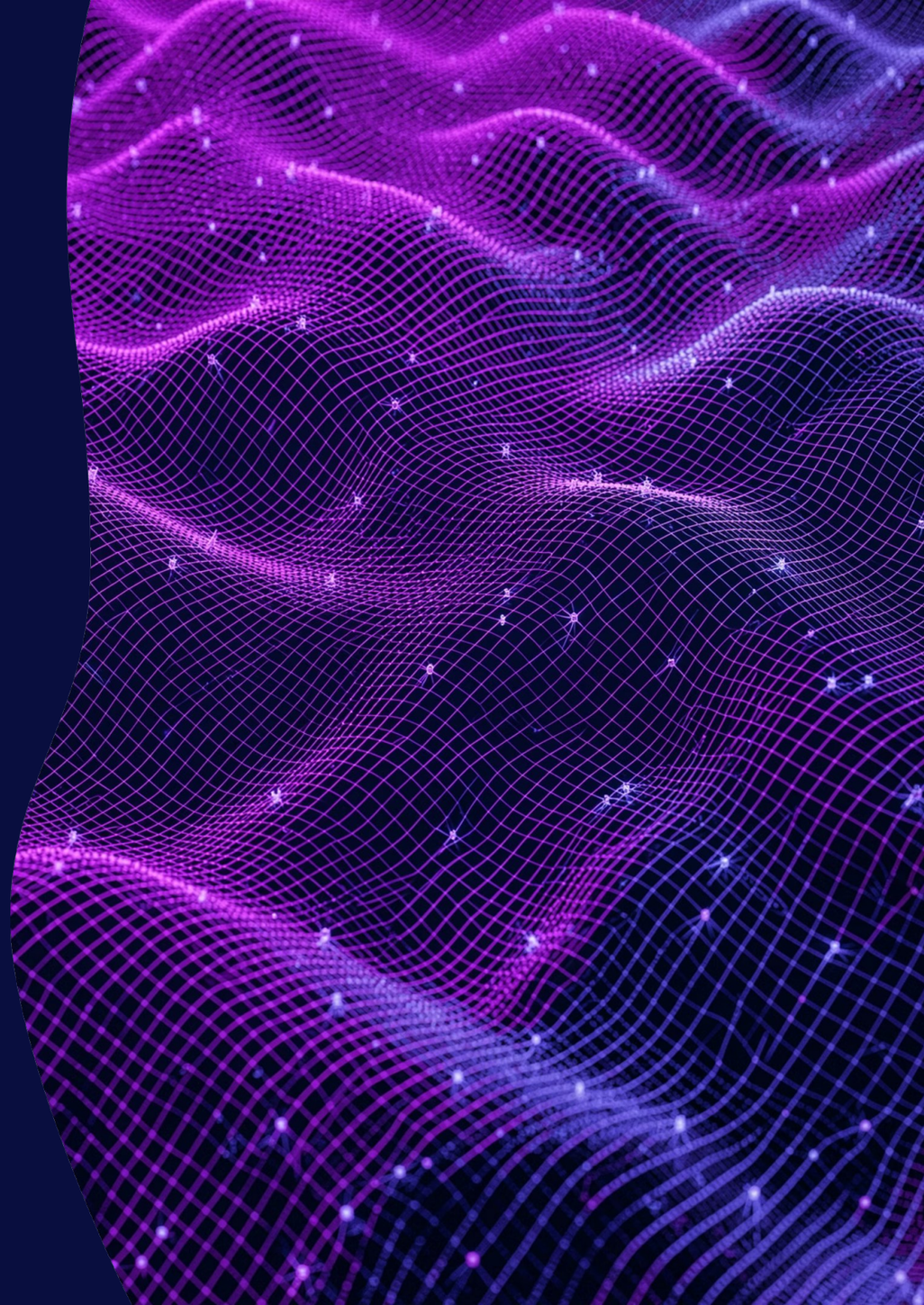
First Conceived

October 2024, SR University — motivated by live Mirai botnet activity on campus network.

Disclosed

NCETC-2025 conference paper; NDA with SR University TTO executed Jan 15, 2026 (SRU-NDA-2026-003).

Filing Recommended

Provisional application to secure priority date; PCT filing designating USA, EU & Japan advised.

Evidence archived at SR University Research Repository — Ref: SRU/CS/2025/BOTNET/001. DST-CRG/2025 funding proposal under preparation.